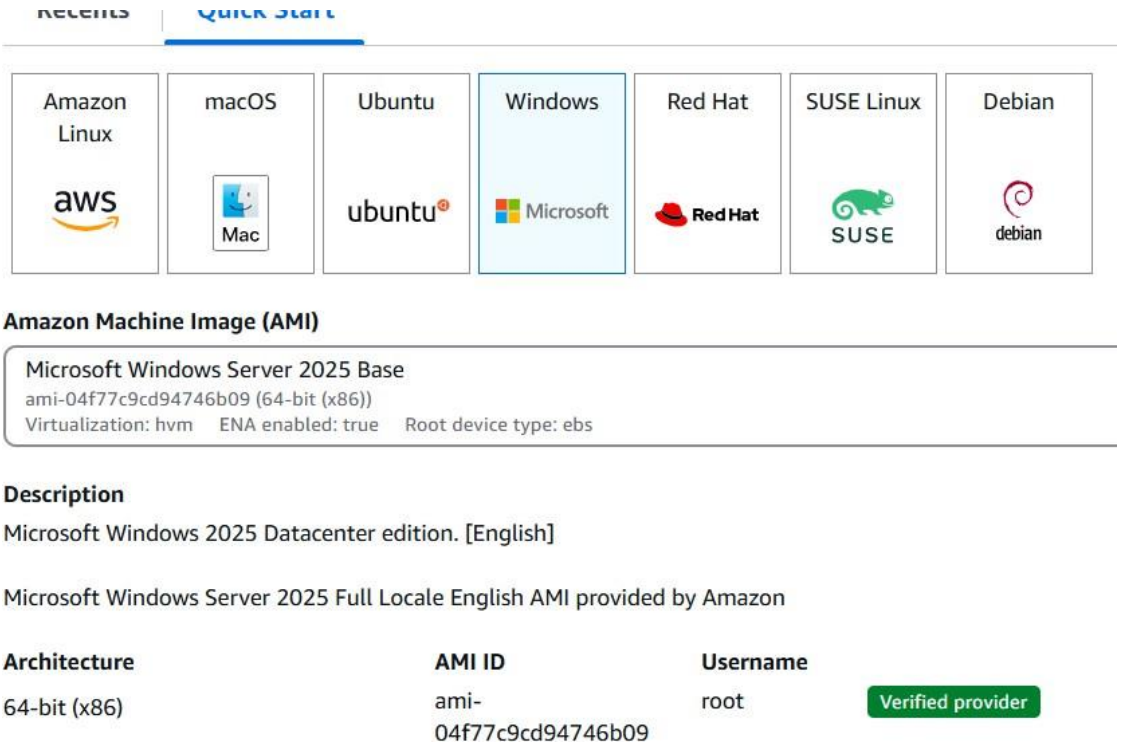
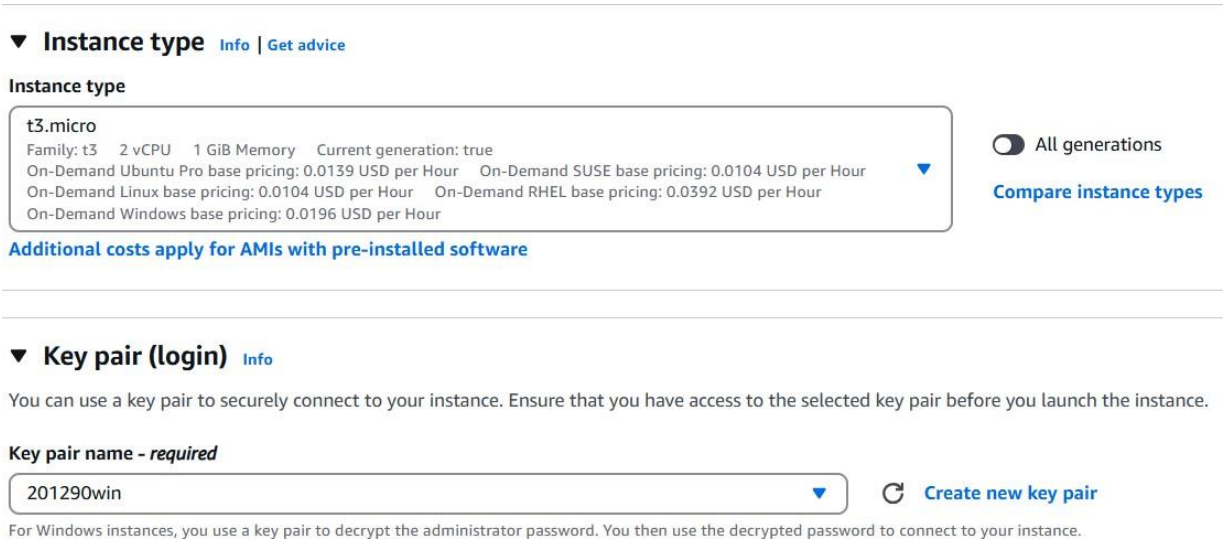
TASK 4

1. How do you launch the server using AMI(Amazon Machine Image)?

→ Step 1: Login to the AWS console
→ Step 2: Click on Launch instance from EC2 dashboard
→ Step 3: Under the Images, Choose AMI (windows, Linux, ubuntu)



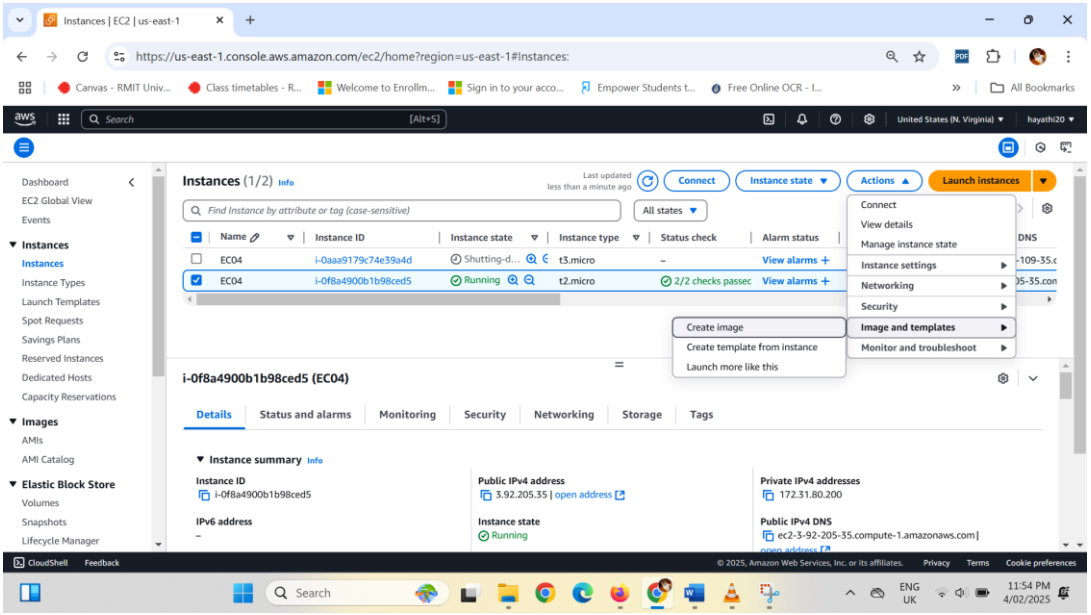
→ Step 4: Choose Instance type (t2. micro) and click on next



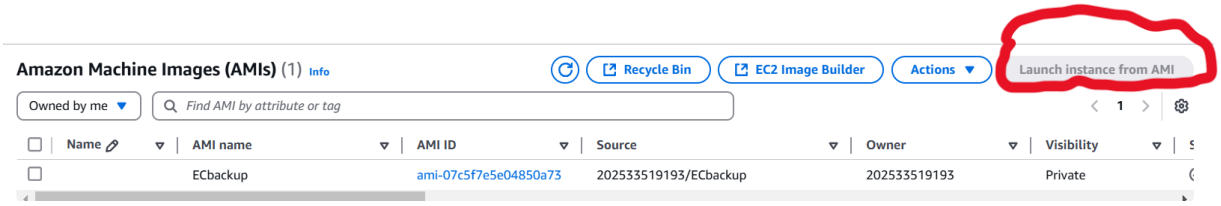
→ Step 5: Configure EBS and security group if needed
→ Step 6: Launch the instance and successfully launched as shown below.



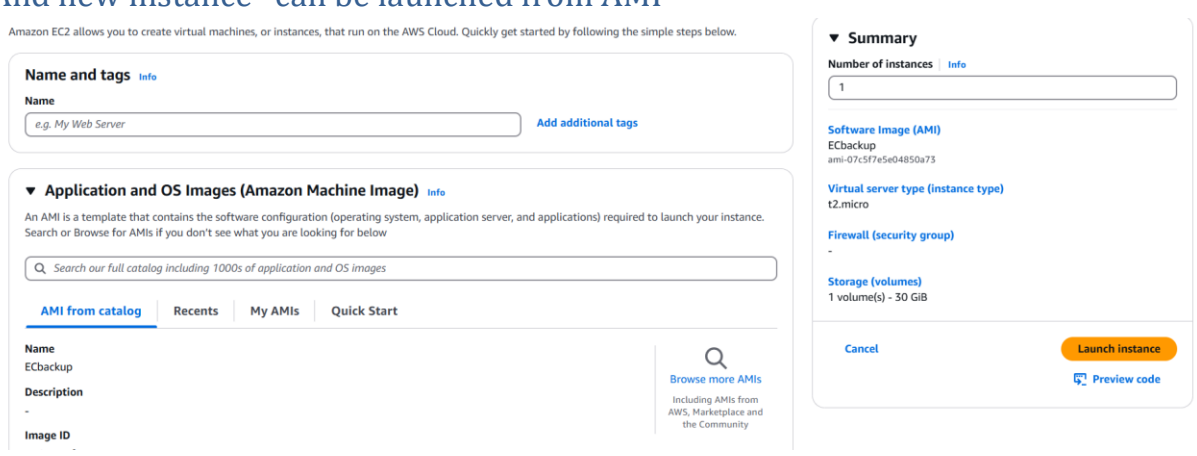
➔ Go to EC2 Dashboard → Instances
➔ Select the instance → Click "Actions" → "Create Image"



Below is the backup server created form EC04 instance.

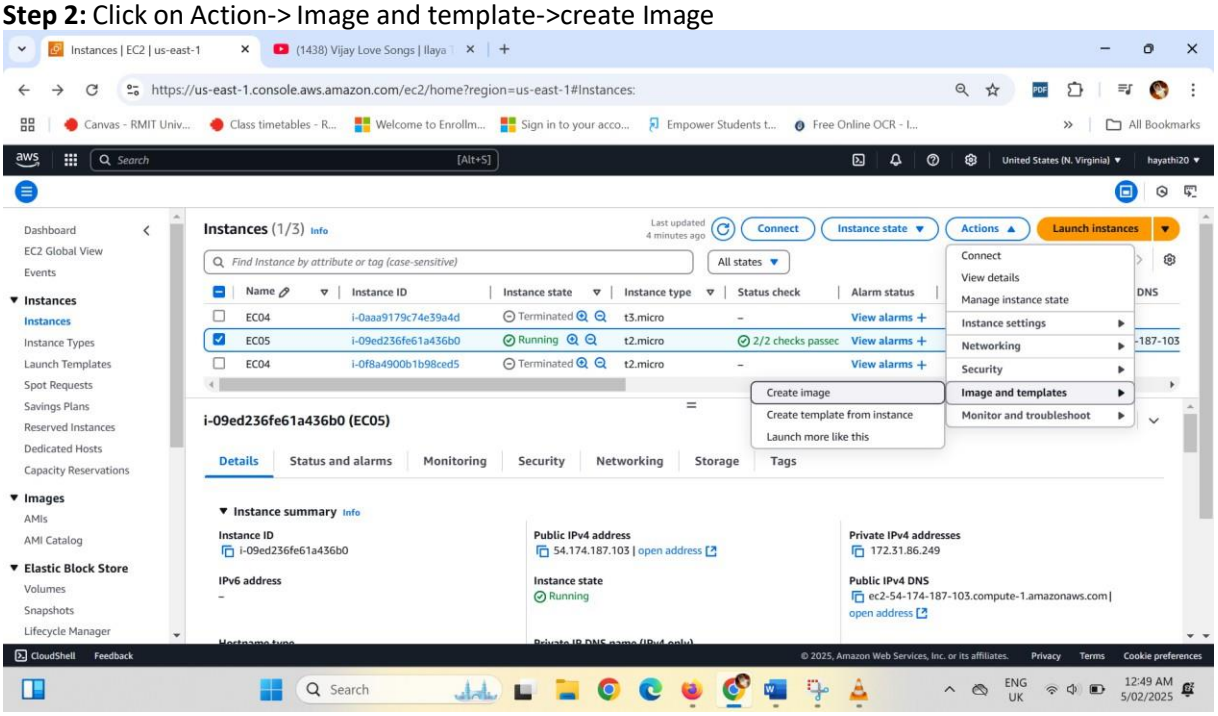


And new instance can be launched from AMI

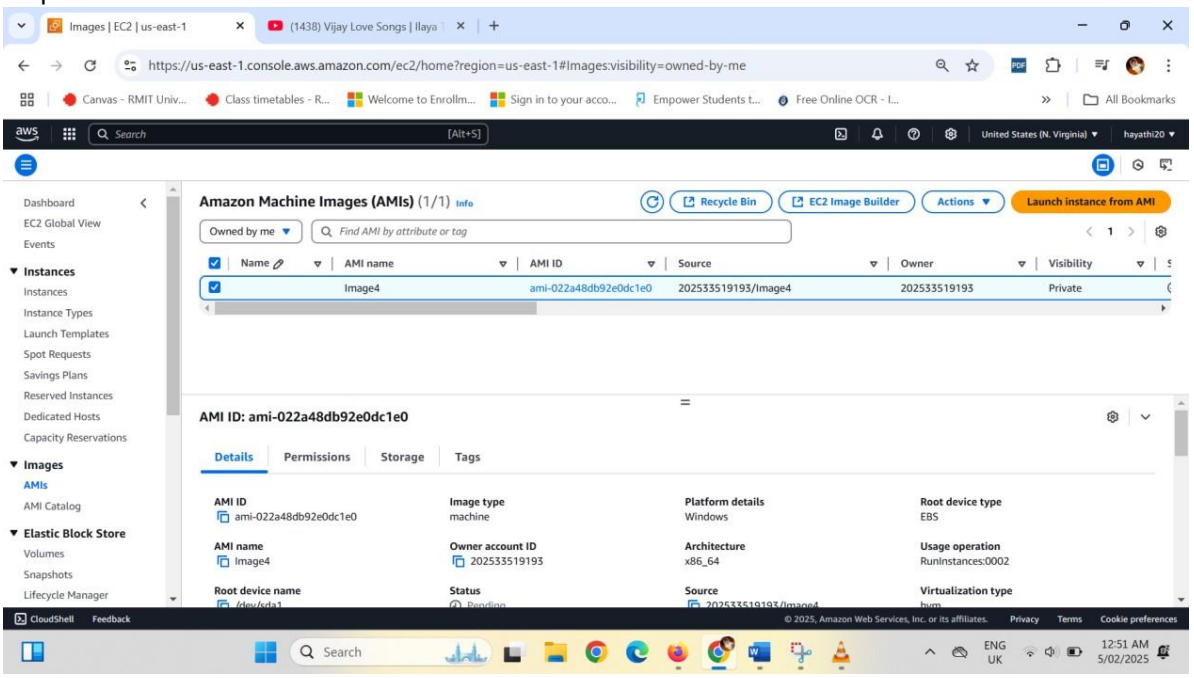


2. How to login to server if you lost your keypair ?

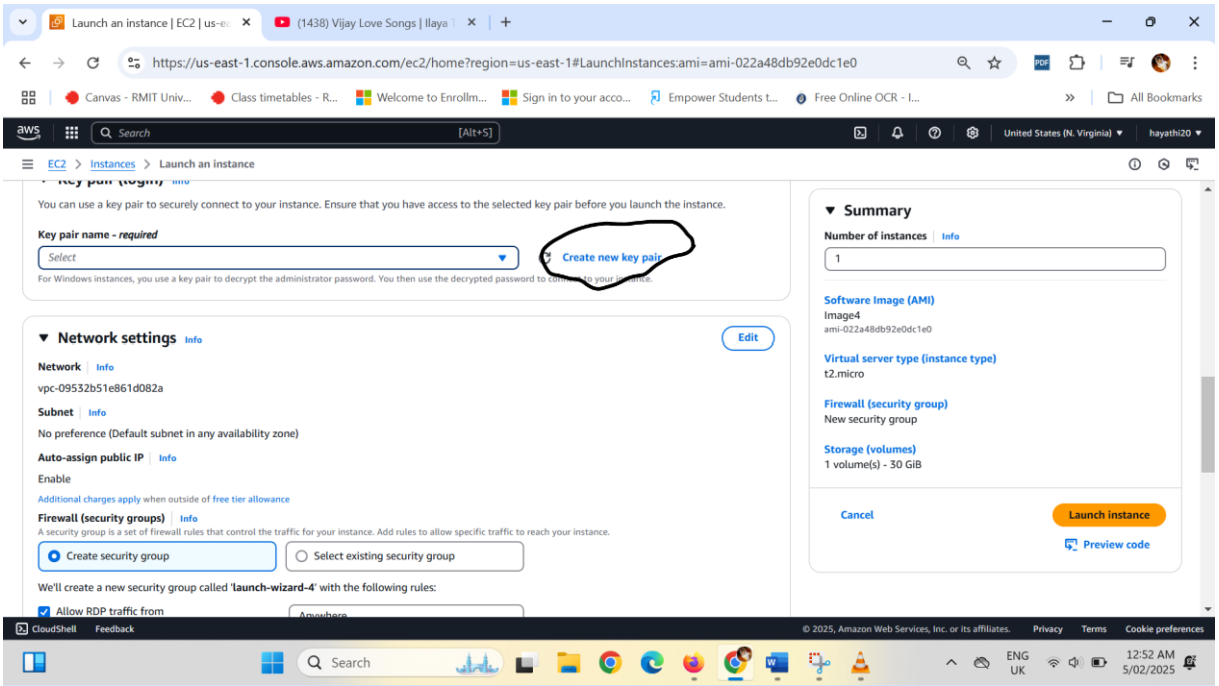
Step 1: By creating EC2 as AMI we can launch new instance and create a new pair key from AMI .
Step 2: Click on Action-> Image and template->create Image

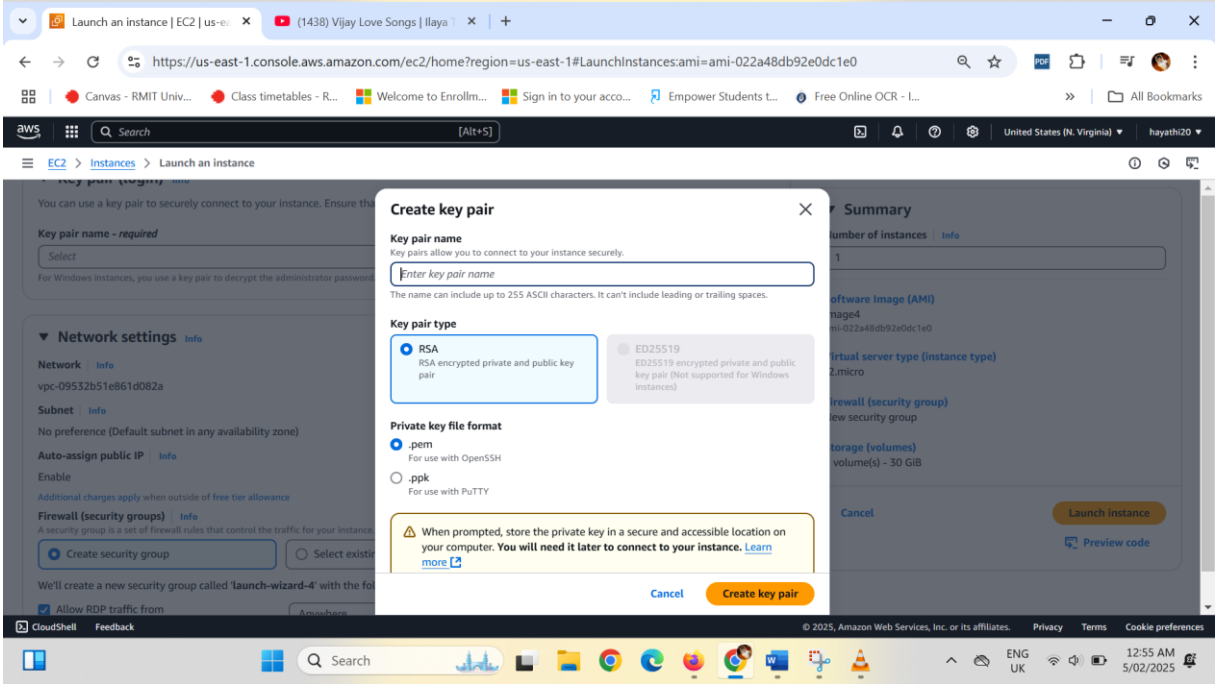


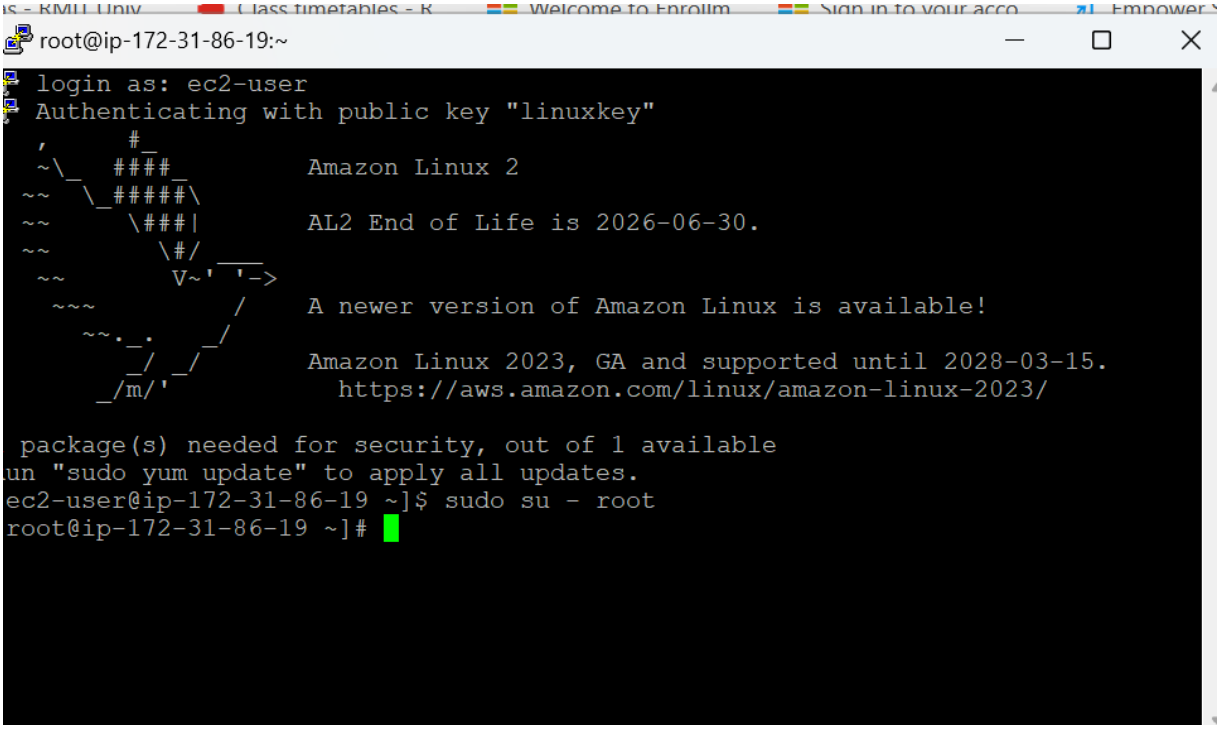
Step3: Click Launch instance from AMI



Step 4: Now select create a key pair to create new and save it in private







Connected with putty ……..

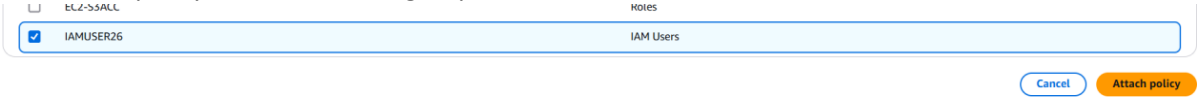
3. How to provide region level restriction for iam users..?

- Go to AWS IAM Console → Policies → Create Policy.
- Select JSON Editor and add the policy
- Mention the regions which the user can be restricted.

```
"Effect": "Deny",
"Action": "*",
"Resource": "*",
"Condition": {
  "StringNotEqualsIfExists": {
    "aws:RequestedRegion": [
      "us-east-2",
      "ap-south-1"
```

Here I have restricted Mumbai and Ohio region for this user.

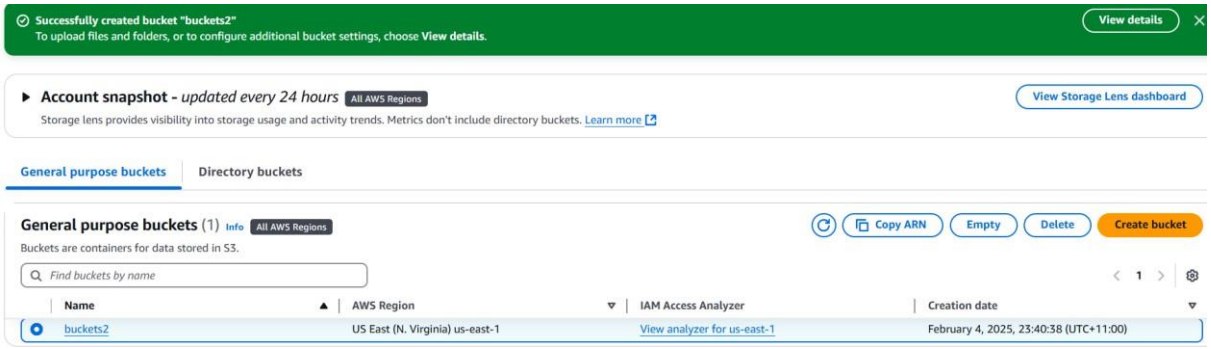
- Attach this policy to the user or in groups.



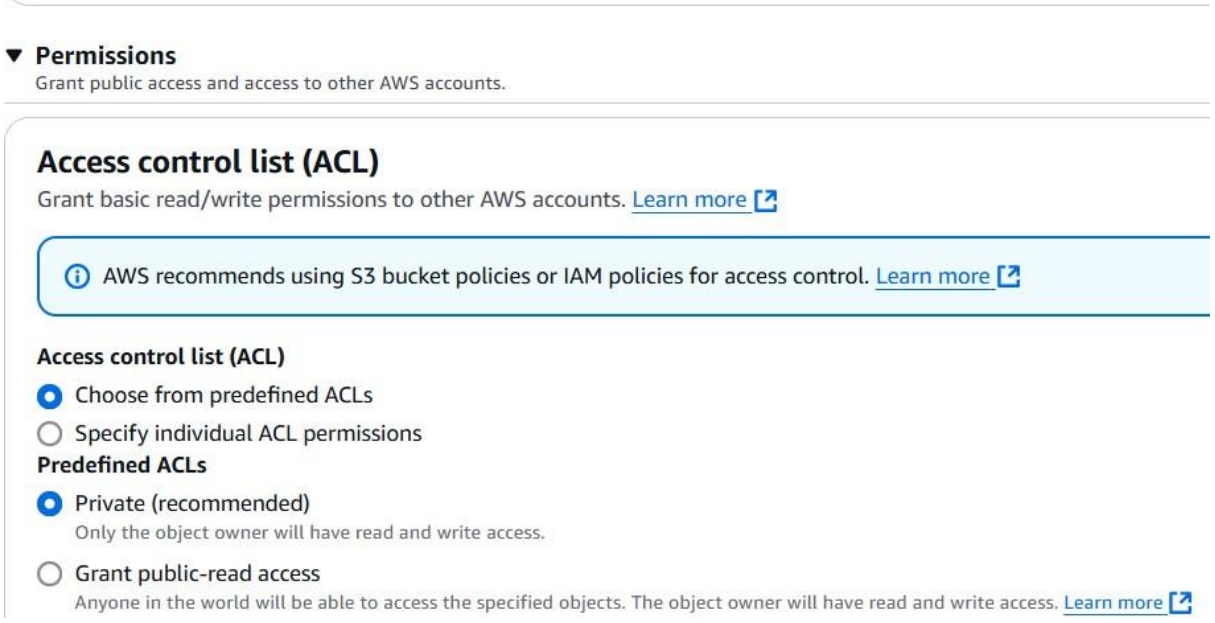
- Select the created policy and save
- 

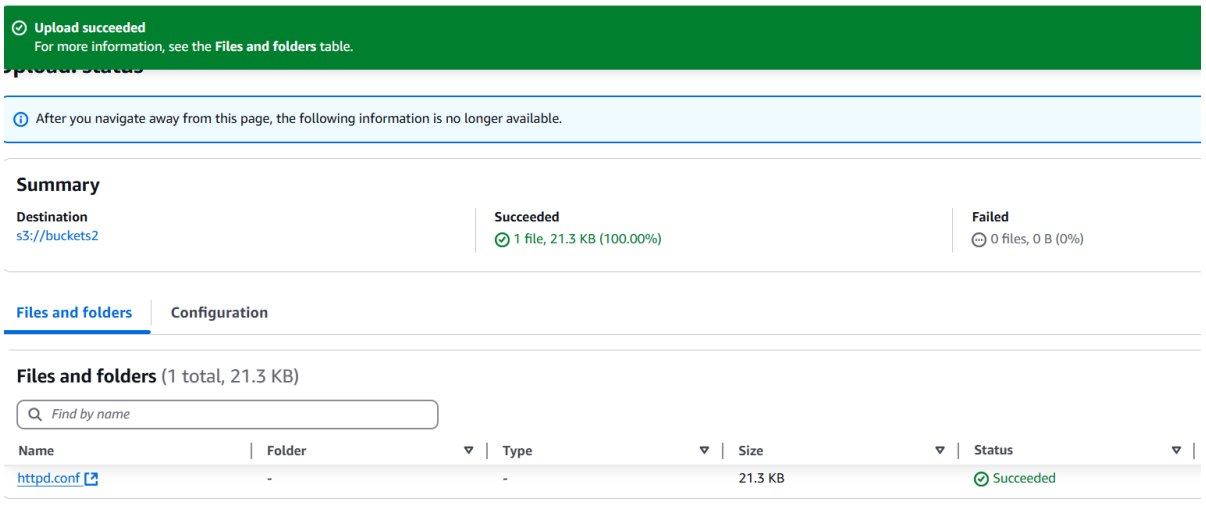
4. How to access object in private bucket.?

Step 1 : Login to AWS console-> S3 -> create bucket

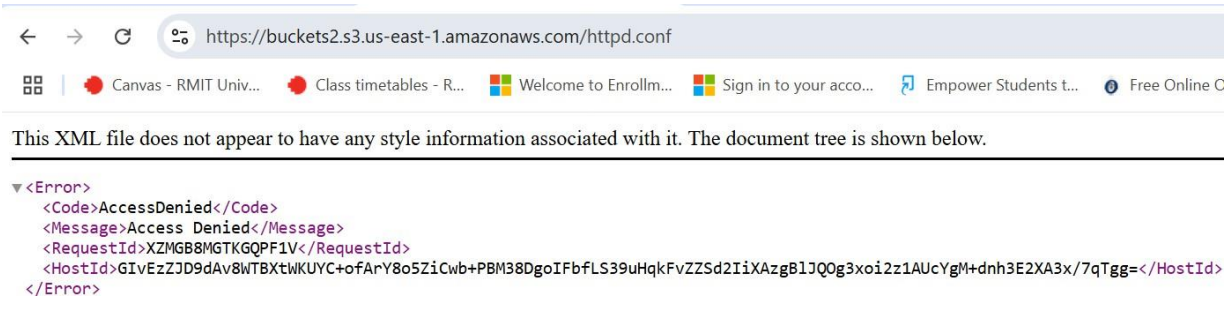


Step 2: Create an object with private access



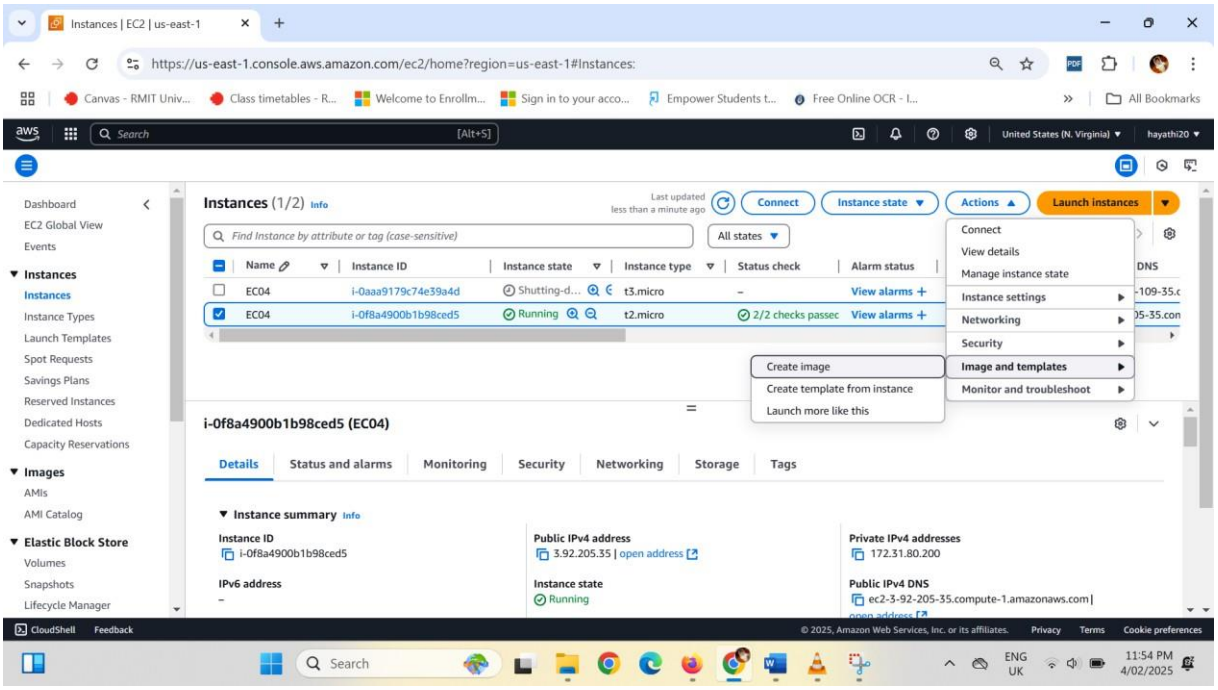


Here I accessed the object in private

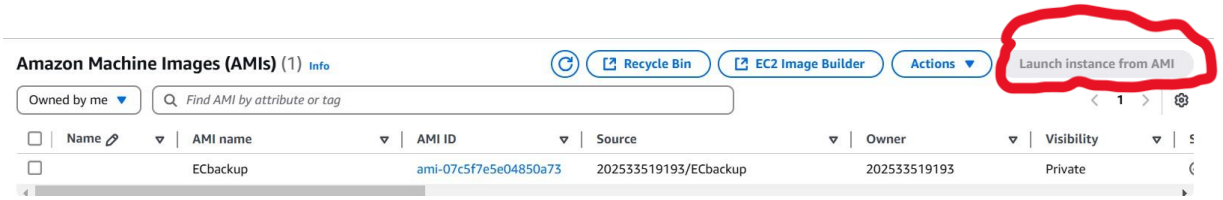


5. How to backup your server?

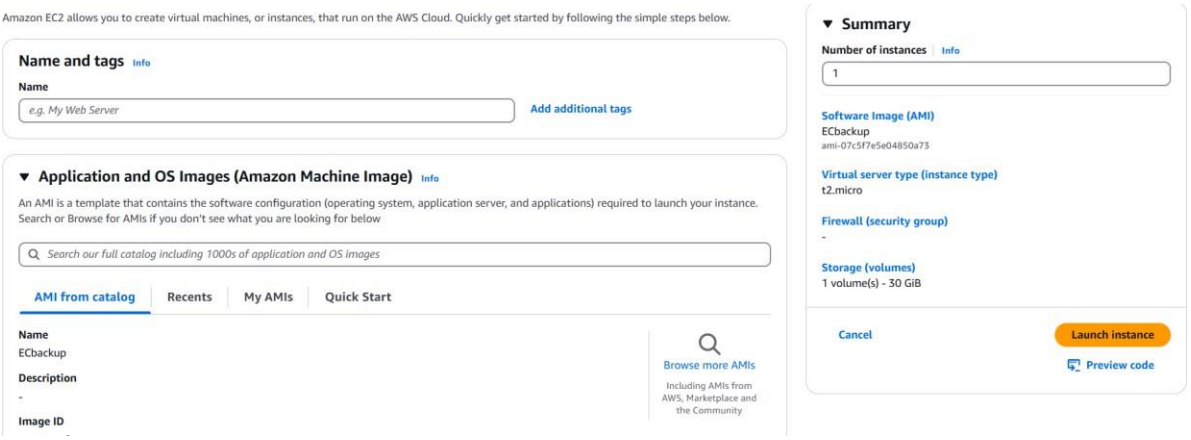
➔ Go to EC2 Dashboard → Instances
➔ Select the instance → Click "Actions" → "Create Image"



Below is the backup server created form EC04 instance.



And new instance can be launched from AMI



6. How to protect your server from all users..?

- Go to the AWS Management Console

  Navigate to EC2 Dashboard → Click on Instances

- Select the Instance

  Click on the instance you want to modify.

- Check the IAM Role Attached

  Scroll down to the IAM Role section in the details pane.

  If a role is attached, it will be listed.

- Detach the IAM Role

  Click Actions → Security → Modify IAM Role

  Select "No Role" or choose None.

  Click Update IAM Role to save changes**.**

7. What is permission boundary in IAM..?

→ It is an IAM policy which sets the maximum permissions in IAM users can have. Users cannot perform outside the defined boundaries.

▼ Set permissions boundary - *optional*

Set a permissions boundary to control the maximum permissions for this user. Use this advanced feature used to delegate permission management to others. Learn more

☑ Use a permissions boundary to control the maximum permissions
You can select one of the existing permissions policies to define the boundary.

Permissions policies (1322)

Select policy to set the permissions boundary.

Search | Filter by Type: All types | 1 2 3 4 5 6 7 ... 67

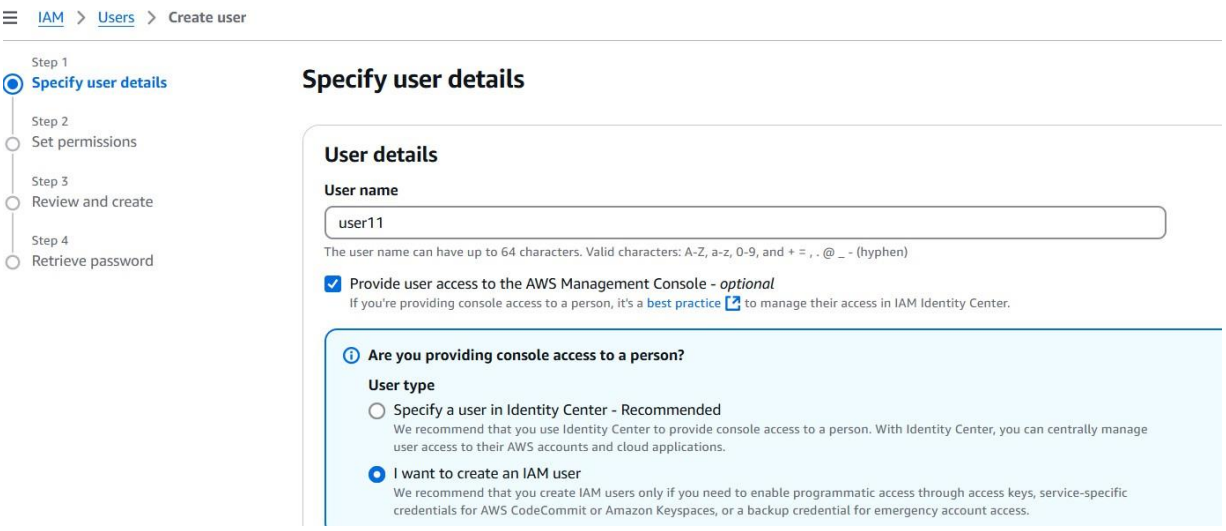
Policy name	Type	Attached entities
AccessAnalyzerServiceRolePolicy	AWS managed	0
AdministratorAccess	AWS managed - job function	0
AdministratorAccess-Amplify	AWS managed	0
AdministratorAccess-AWSElasticBeanstalk	AWS managed	0
AIOpsAssistantPolicy	AWS managed	0

8. IAM user can change them password...? If yes means how...? If no means why..?

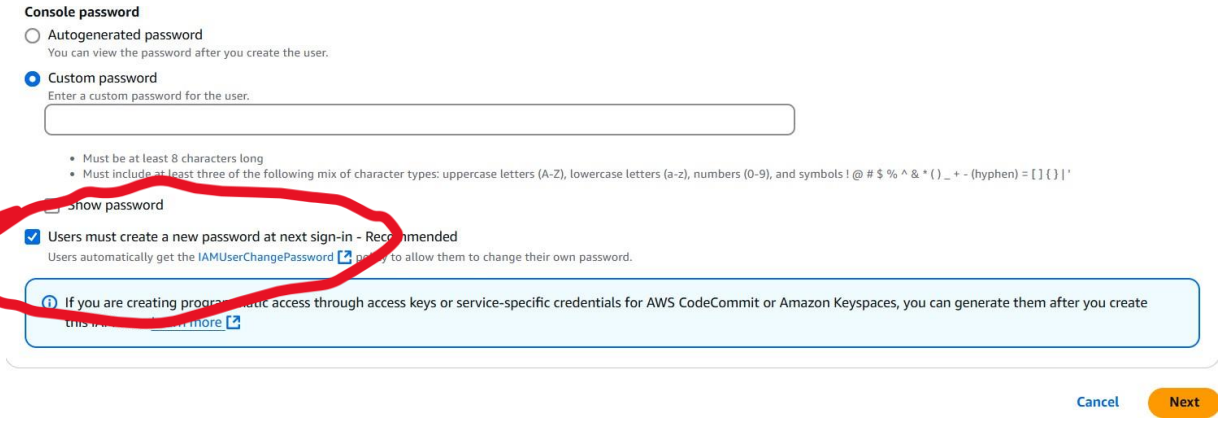
→ If a IAM users have necessary permissions to change the password they could do it. It can be done in following steps

Step 1: Sign in to AWS Management console

Step 2: Go to -> IAM->Users -> create user



Step 3: Under console password-> select any one option and enable the highlighted section. So user can change the password when the login for first time. If this section not ticked user cannot change the password for the account.



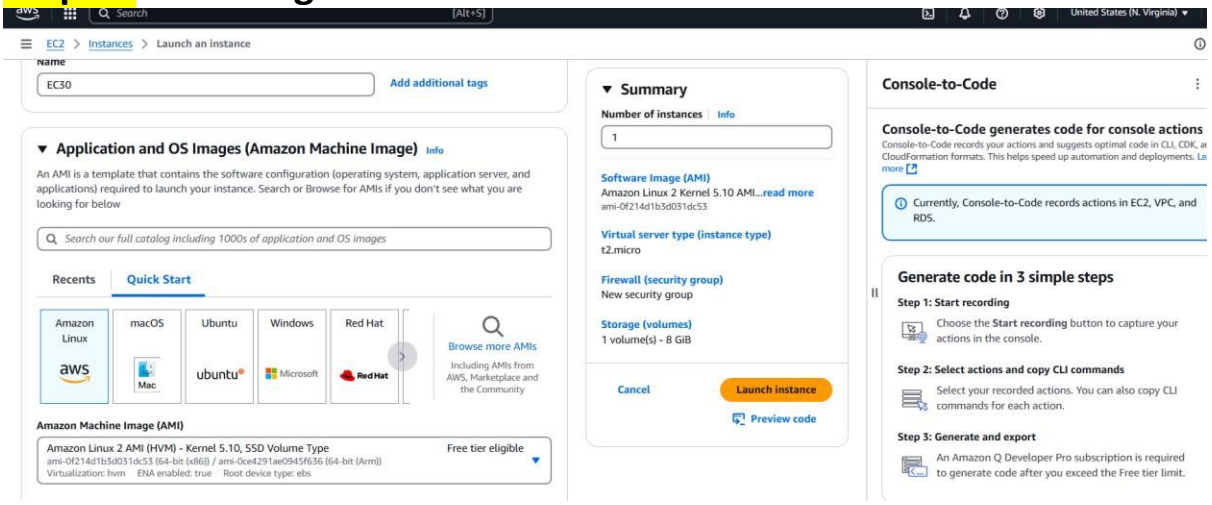
Step 4: Click on next and create an IAM user account

9. How many instances can create at the time..?

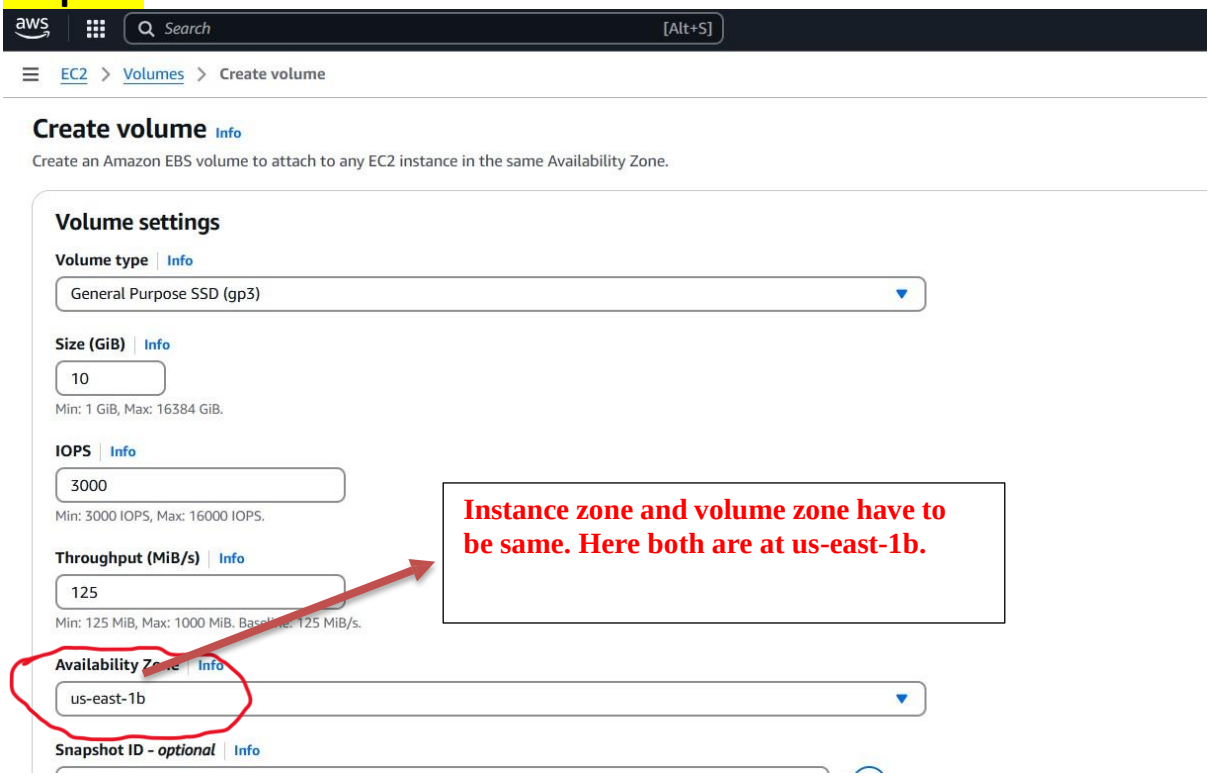
→ For new accounts it is limited to 20 instances per region .

10. How to mount volume in linux instance...?

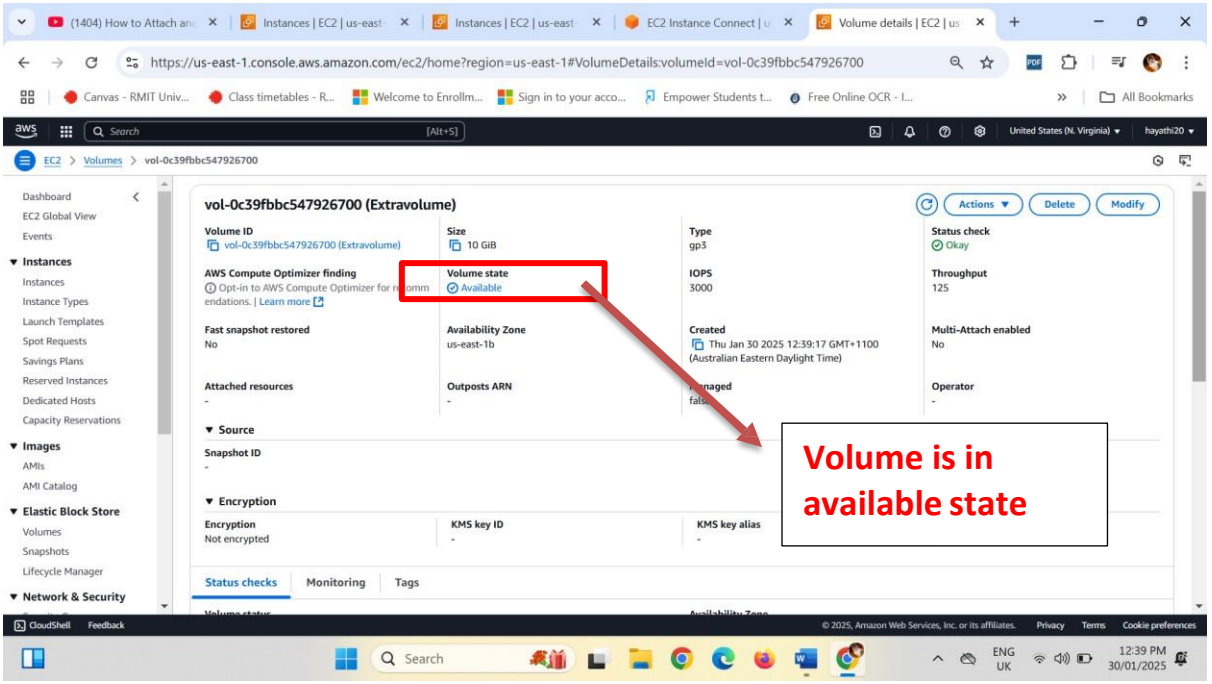
Step 1: Launching a new Instance



Step 2 : To Create a volume Goto-> Elastic Block store-> Volume

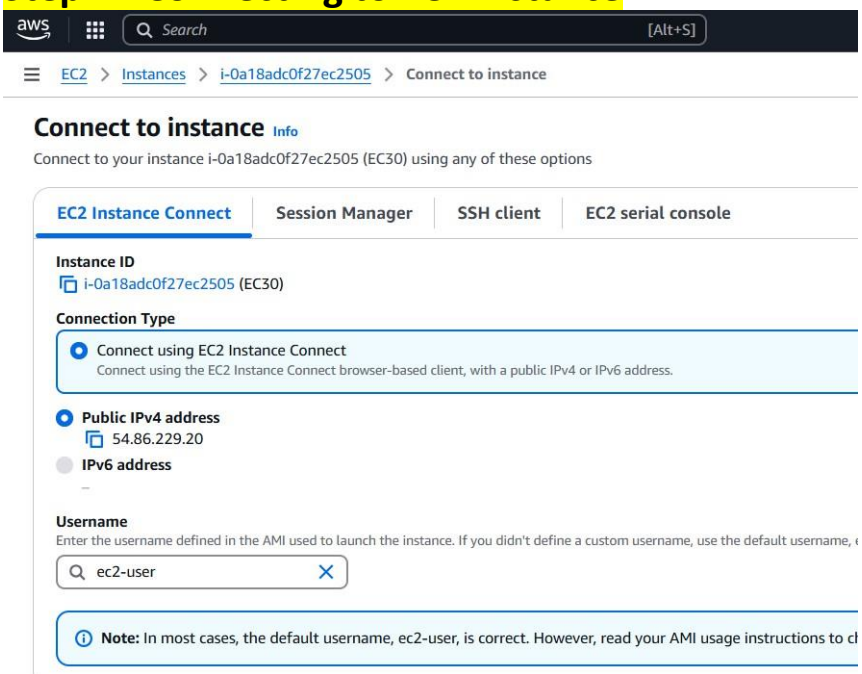


Step 3: Click on Create volume and a new volume is created.

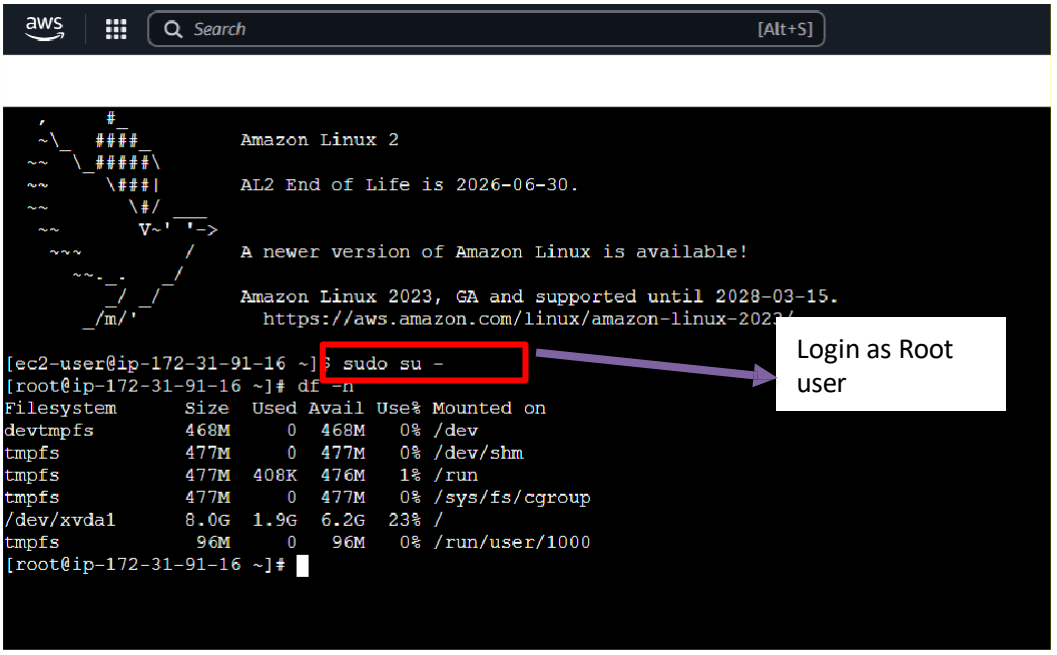


The next step is to connect this available volume with the Instance.

Step 4 : Connecting to EC2 Instance



After successful connection, Amazon Linux is opened.

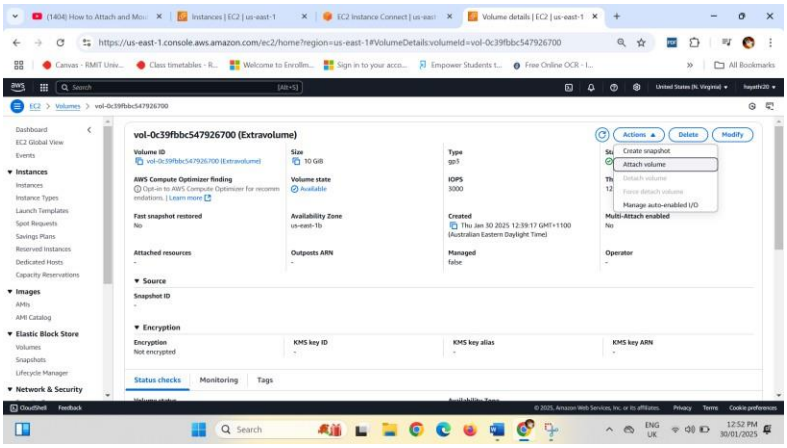


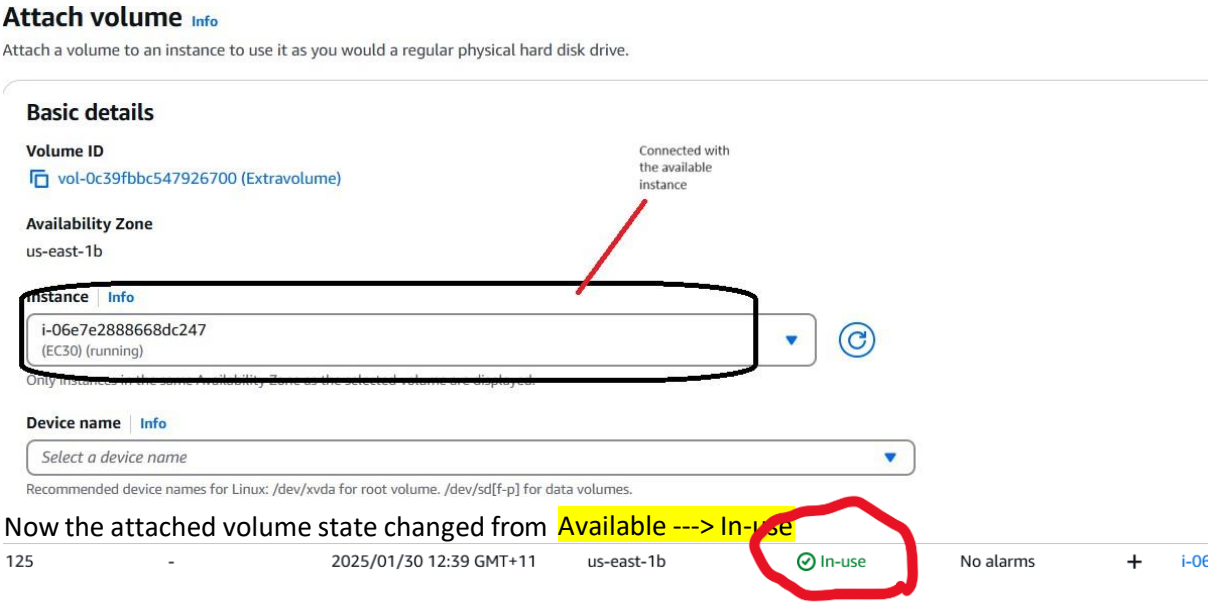
Use **df -h** command to check available volume connected with our instance.

```
[root@ip-172-31-91-16 ~]# df -h
Filesystem      Size  Used Avail Use% Mounted on
devtmpfs        468M     0  468M   0% /dev
tmpfs           477M     0  477M   0% /dev/shm
tmpfs           477M  408K  476M   1% /run
tmpfs           477M     0  477M   0% /sys/fs/cgroup
/dev/xvda1      8.0G  1.9G  6.2G  23% /
tmpfs            96M     0   96M   0% /run/user/1000
[root@ip-172-31-91-16 ~]#
```

Step 5 : Attach the volume with the Instance

Goto Volume-> Select Attach volume



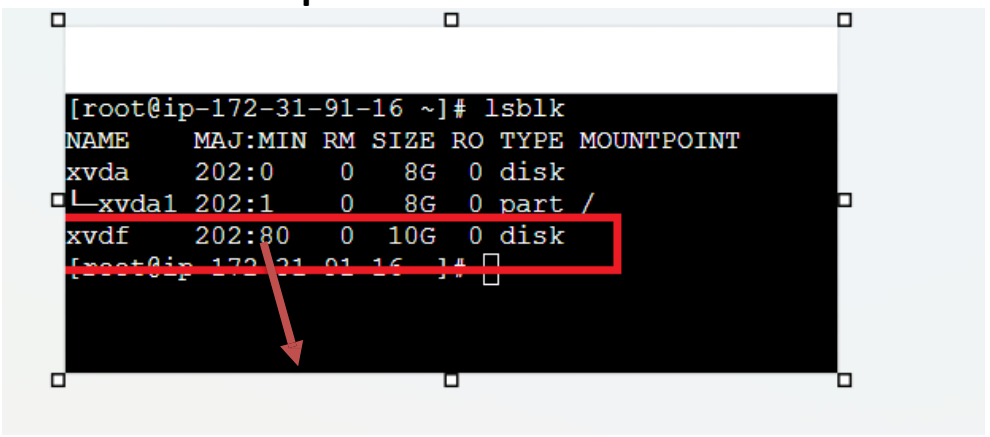


Step 6: Then , Checking in EC2 Instance if it is showing under the EC30 Instance

```
[root@ip-172-31-91-16 ~]# df -h
Filesystem      Size  Used Avail Use% Mounted on
devtmpfs        468M     0  468M   0% /dev
tmpfs           477M     0  477M   0% /dev/shm
tmpfs           477M  412K  476M   1% /run
tmpfs           477M     0  477M   0% /sys/fs/cgroup
/dev/xvda1      8.0G  1.9G  6.2G  23% /
tmpfs            96M     0   96M   0% /run/user/1000
[root@ip-172-31-91-16 ~]#
```

It is still not displayed here because we need to create a directory and mount the EBS volume to the Instance.

Step 7: Use the lsblk command to view your available disk devices and their mount points



Attached volume can now viewed in the Instance (EC30)

Step 8: Check if there is any file system on new EBS Volume:

file -s /dev/xvdf (If you see "Data", meaning you need to setup file system for this block device.)

```
xvdf    202:80   0  10G  0 disk
[root@ip-172-31-91-16 ~]# file -s /dev/xvdf
/dev/xvdf: data
[root@ip-172-31-91-16 ~]#
```

Setting up file system:

1. Create a file system on volume to mount it to EC2:

mkfs -t xfs /dev/xvdf

```
[root@ip-172-31-91-16 ~]# mkfs -t xfs /dev/xvdf
meta-data=/dev/xvdf              isize=512    agcount=4, agsize=655360 blks
         =                       sectsz=512   attr=2, projid32bit=1
         =                       crc=1        finobt=1, sparse=1, rmapbt=0
         =                       reflink=1    bigtime=0 inobtcount=0
data     =                       bsize=4096   blocks=2621440, imaxpct=25
         =                       sunit=0      swidth=0 blks
naming   =version 2              bsize=4096   ascii-ci=0, ftype=1
log      =internal log           bsize=4096   blocks=2560, version=2
         =                       sectsz=512   sunit=0 blks, lazy-count=1
realtime =none                   extsz=4096   blocks=0, rtextents=0
[root@ip-172-31-91-16 ~]#
```

2. Create new directory:

mkdir -p /apps/my-data/apps/volume/new-volume
cd /apps/my-data/

```
[root@ip-172-31-91-16 ~]# mkdir -p /apps/my-data/apps/volume/new-volume
[root@ip-172-31-91-16 ~]#  cd /apps/my-data/
[root@ip-172-31-91-16 my-data]#
```

Directory is created.

Step 9: Mount the volume

mount /dev/xvdf /apps/my-data/apps/volume/new-volume

```
[root@ip-172-31-91-16 ~]# lsblk
NAME    MAJ:MIN RM SIZE RO TYPE MOUNTPOINT
xvda    202:0    0   8G  0 disk
└─xvda1 202:1    0   8G  0 part /
xvdf    202:80   0  10G  0 disk
[root@ip-172-31-91-16 ~]# file -s /dev/xvdf
/dev/xvdf: data
[root@ip-172-31-91-16 ~]# mkfs -t xfs /dev/xvdf
meta-data=/dev/xvdf              isize=512    agcount=4, agsize=655360 blks
         =                       sectsz=512   attr=2, projid32bit=1
         =                       crc=1        finobt=1, sparse=1, rmapbt=0
         =                       reflink=1    bigtime=0 inobtcount=0
data     =                       bsize=4096   blocks=2621440, imaxpct=25
         =                       sunit=0      swidth=0 blks
naming   =version 2              bsize=4096   ascii-ci=0, ftype=1
log      =internal log           bsize=4096   blocks=2560, version=2
         =                       sectsz=512   sunit=0 blks, lazy-count=1
realtime =none                   extsz=4096   blocks=0, rtextents=0
[root@ip-172-31-91-16 ~]# mkdir -p /apps/my-data/apps/volume/new-volume
[root@ip-172-31-91-16 ~]#  cd /apps/my-data/
[root@ip-172-31-91-16 my-data]# mount /dev/xvdf /apps/my-data/apps/volume/new-volume
[root@ip-172-31-91-16 my-data]#
```

Step 10: To check if the mounting is successful, use df -h command

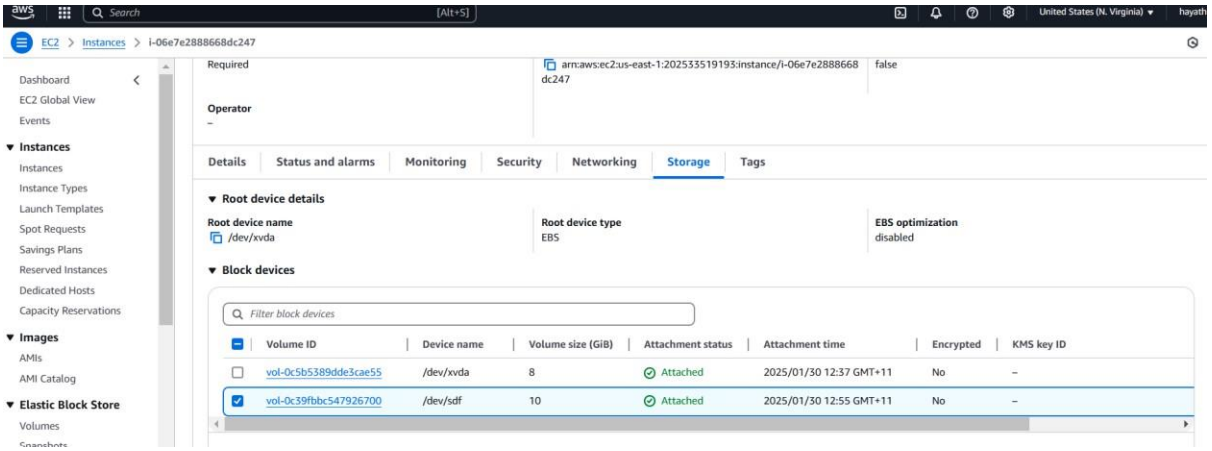
```
realtime =none                   extsz=4096   blocks=0, rtextents=0
[root@ip-172-31-91-16 ~]# mkdir -p /apps/my-data/apps/volume/new-volume
[root@ip-172-31-91-16 ~]#  cd /apps/my-data/
[root@ip-172-31-91-16 my-data]# mount /dev/xvdf /apps/my-data/apps/volume/new-volume
[root@ip-172-31-91-16 my-data]# df -h
Filesystem      Size  Used Avail Use% Mounted on
devtmpfs        468M     0  468M   0% /dev
tmpfs           477M     0  477M   0% /dev/shm
tmpfs           477M  468K  476M   1% /run
tmpfs           477M     0  477M   0% /sys/fs/cgroup
/dev/xvda1      8.0G  1.9G  6.2G  23% /
tmpfs            96M     0   96M   0% /run/user/1000
tmpfs            96M     0   96M   0% /run/user/0
/dev/xvdf        10G  104M  9.9G   2% /apps/my-data/apps/volume/new-volume
```

Mounting is successful and it is available in Instance (EC30)

```
[root@ip-172-31-91-16 my-data]# lsblk
NAME    MAJ:MIN RM SIZE RO TYPE MOUNTPOINT
xvda    202:0    0   8G  0 disk
└─xvda1 202:1    0   8G  0 part /
xvdf    202:80   0  10G  0 disk /apps/my-data/apps/volume/new-volume
```

Previously it was not listed, and it is available now after successful mounting.

Also, it could be confirmed by viewing Storage in Particular Instance in AWS console.


aws
Search
[Alt+S]
United States (N. Virginia)
hayath
EC2 > Instances > i-06e7e2888668dc247
Dashboard
EC2 Global View
Events
Instances
Instances
Instance Types
Launch Templates
Spot Requests
Savings Plans
Reserved Instances
Dedicated Hosts
Capacity Reservations
Images
AMIs
AMI Catalog
Elastic Block Store
Volumes
Required
arn:aws:ec2:us-east-1:202533519193:instance/i-06e7e2888668dc247
false
Operator
–
Details
Status and alarms
Monitoring
Security
Networking
Storage
Tags
Root device details
Root device name
/dev/xvda
Root device type
EBS
EBS optimization
disabled
Block devices
Filter block devices
Volume ID
Device name
Volume size (GiB)
Attachment status
Attachment time
Encrypted
KMS key ID
vol-0c5b5389dde3cae55
/dev/xvda
8
Attached
2025/01/30 12:37 GMT+11
No
–
vol-0c39fbbc547926700
/dev/sdf
10
Attached
2025/01/30 12:55 GMT+11
No
–

→ Create new directory:

mkdir -p /apps/my-data/apps/volume/new-volume
cd /apps/my-data/

→ **Mount the volume**

mount /dev/xvdf /apps/my-data/apps/volume/new-volume

➔ To check if the mounting is successful, use df -h command